

# Generating Rail Scenes Images for Semantic Segmentation using Text-to-Image Diffusion Models

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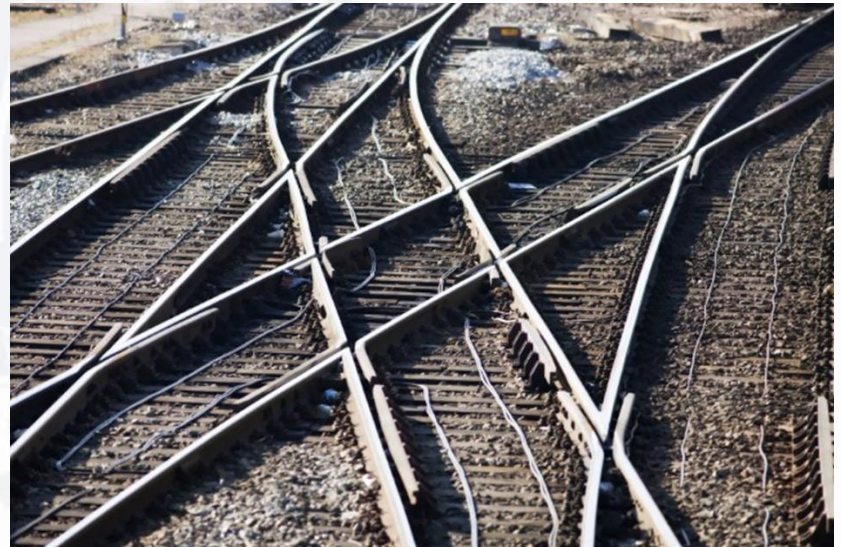
proiect de cercetare realizat alături de Stud. Alexandrescu Andrei-Robert și Stud.  
Drd. Manole Alexandru sub supervizarea doamnei Prof. Dr. Dioșan Laura

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2. Approaches
3. First Steps
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# Context

- Autonomous trains must understand complex rail scenarios
- Some scenarios are rare and not well represented in datasets
- Large datasets are a must
- Collecting images with rare scenarios => unfeasible

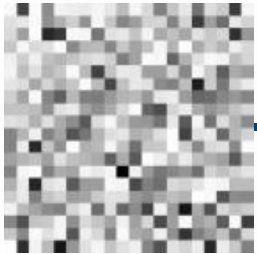


**Solution: Generate realistic synthetic scenarios**

# Approaches

- Use Stable Diffusion (State-of-The-Art in Image Generation)

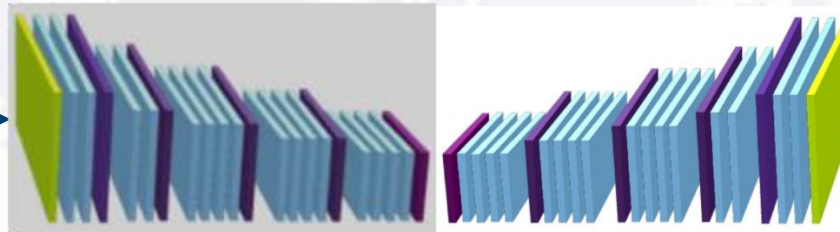
Noise Image



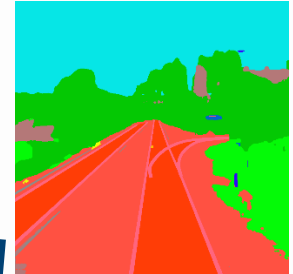
Prompt

Ex: "High-quality Railway Image with a switch going right"

Network



Segmentation mask



Generated Image



Real Image



How realistic is it?



Scores (FID)

# First Steps

- Understanding Stable Diffusion and Control Nets:
  - Literature Review
  - Understand how Stable Diffusion works
  - Validate models on basic tasks



Input Canny edge



Default



"masterpiece of fairy tale, giant deer, golden antlers"



"..., quaint city Galic"

Input

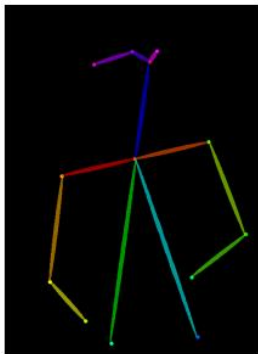
Output

Zhang, Lvmin, Anyi Rao, and Maneesh Agrawala. "Adding conditional control to text-to-image diffusion models." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.

8/14/2024

# First Steps

- Understanding Stable Diffusion and Control Nets:
  - Literature Review
  - Understand how Stable Diffusion works
  - Validate models on basic tasks



Input human pose



Default



"chef in kitchen"



"Lincoln statue"

Input

Output

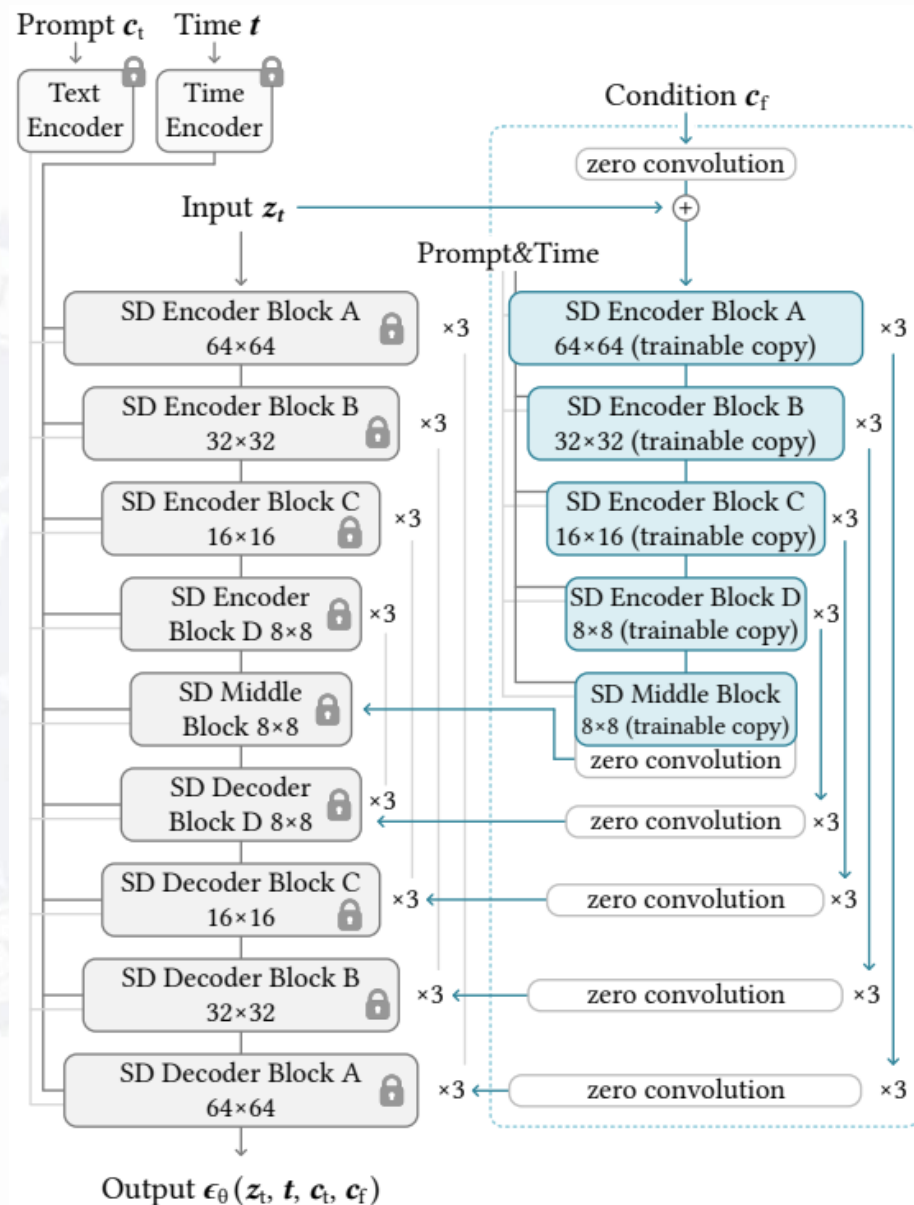
Zhang, Lvmin, Anyi Rao, and Maneesh Agrawala. "Adding conditional control to text-to-image diffusion models." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.

8/14/2024

# ControlNet

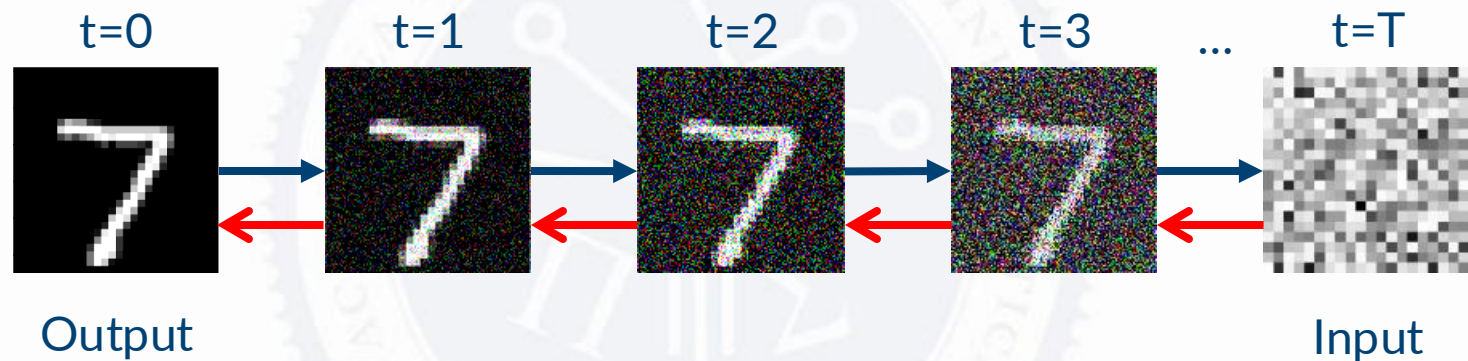
- Recent (Nov. 2023)
- **Idea:** Add more control to Stable Diffusion
- Locked: preserve production-ready model
- Unlocked: allow conditioning
- Zero-convolutions

But what is Stable Diffusion?



# Stable Diffusion

- **Idea:** Progressively add noise and learn how to denoise images

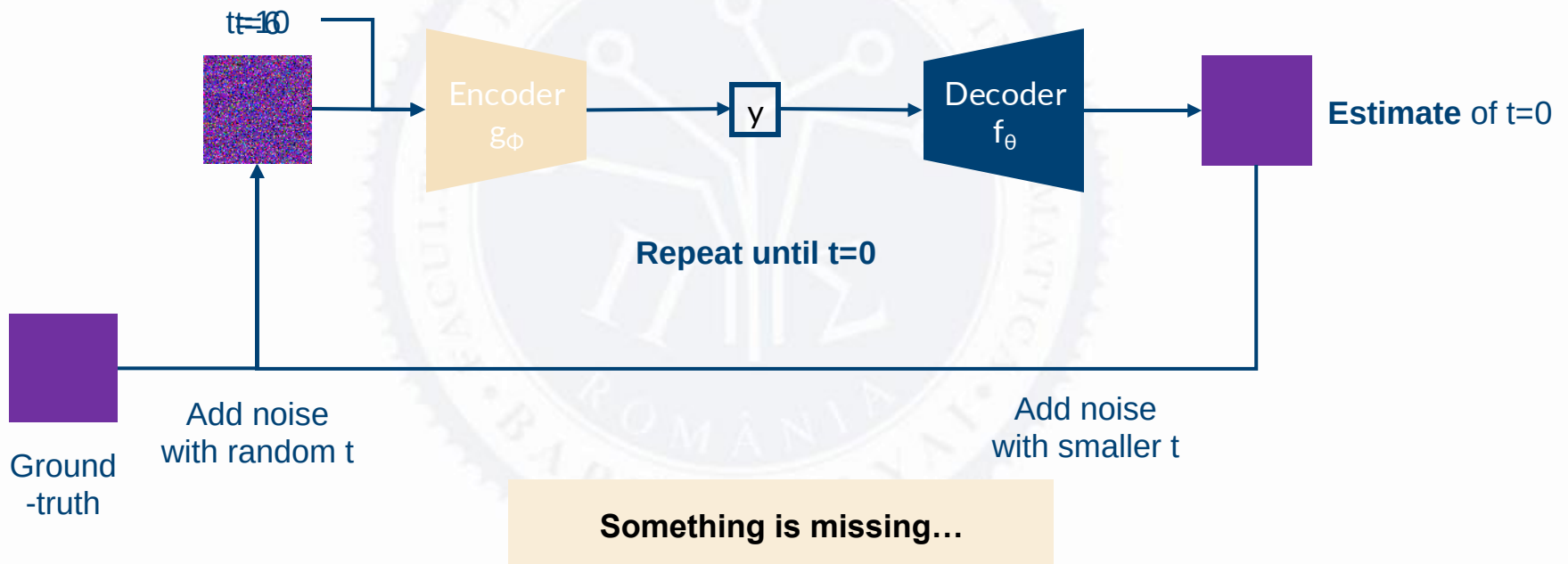


Train a network to undo the process

# Stable Diffusion

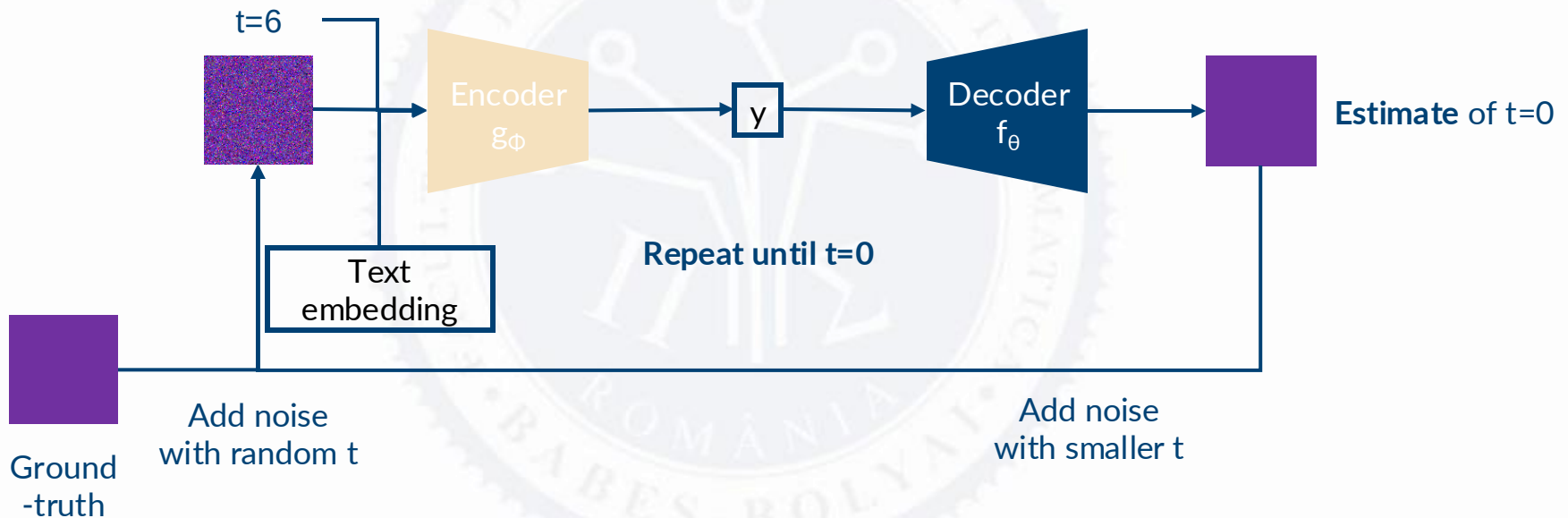
With each repetition, it becomes easier to generate the image

- It is difficult to go from  $t=T$  to  $t=0$

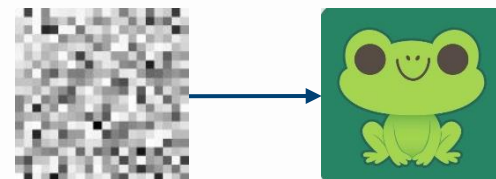


# Stable Diffusion

- Prompts guide the generation, however they are **not exact enough**

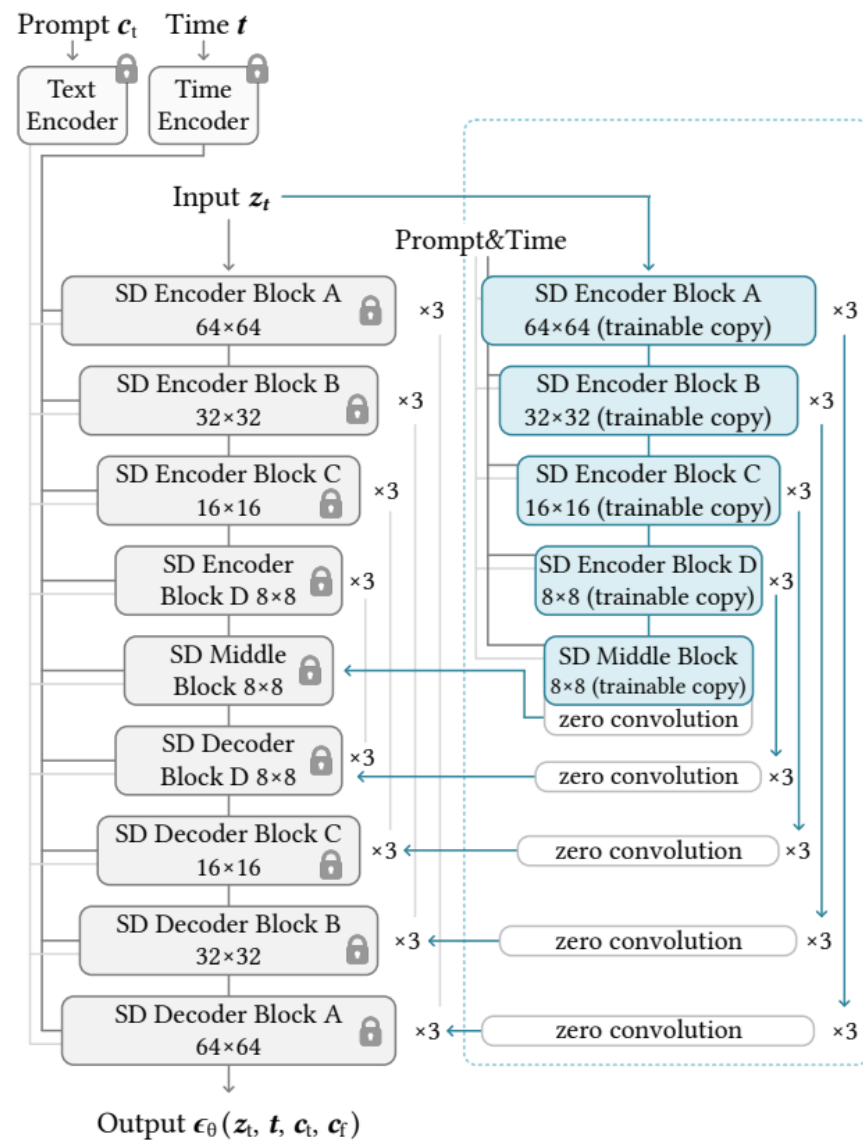


- Inference starts from **full noise image**

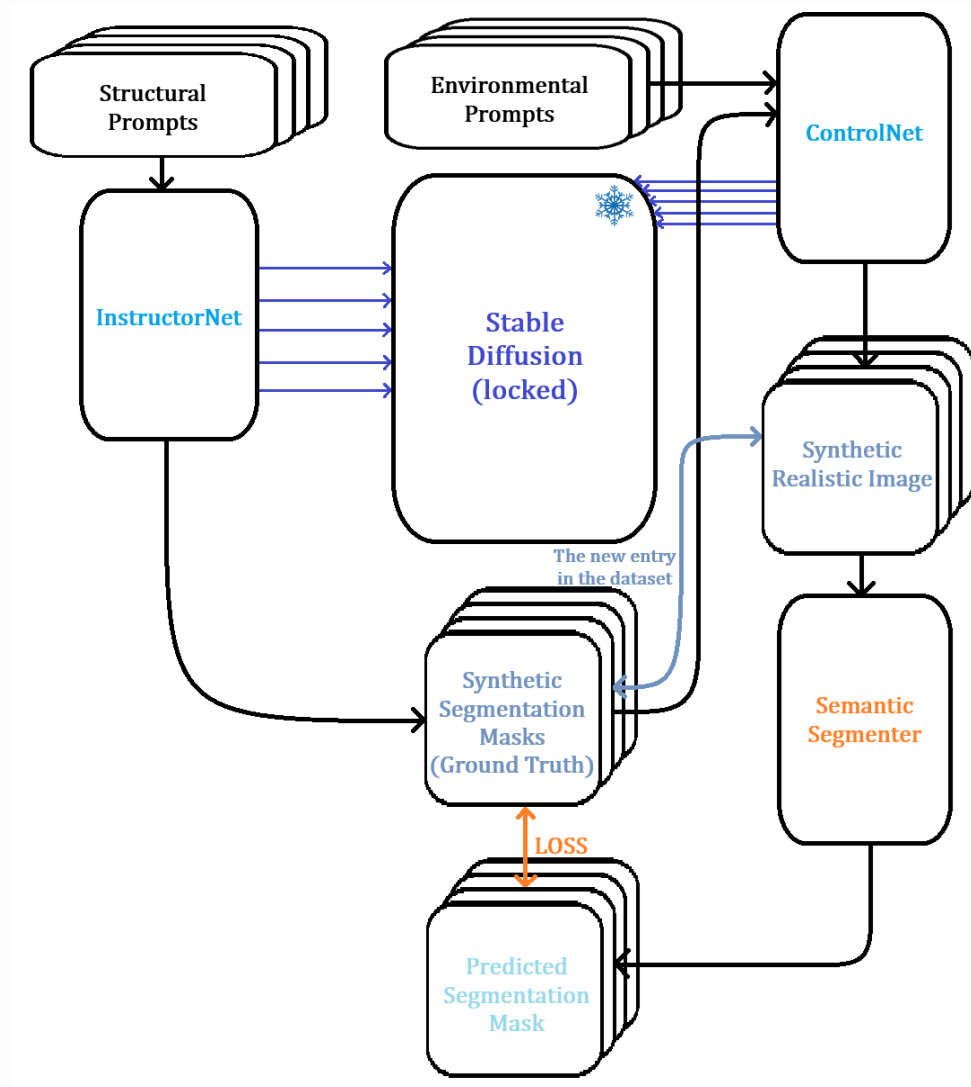


# InstructorNet

- **Idea:** Takes the idea of a HyperNetwork over Stable Diffusion, but without the conditioning, just a way to fine-tune Stable Diffusion without affecting the weights of the production-ready model
- **Why?** As shown in different research articles, just fine-tuning Stable Diffusion leads to "overfitting and catastrophic forgetting".

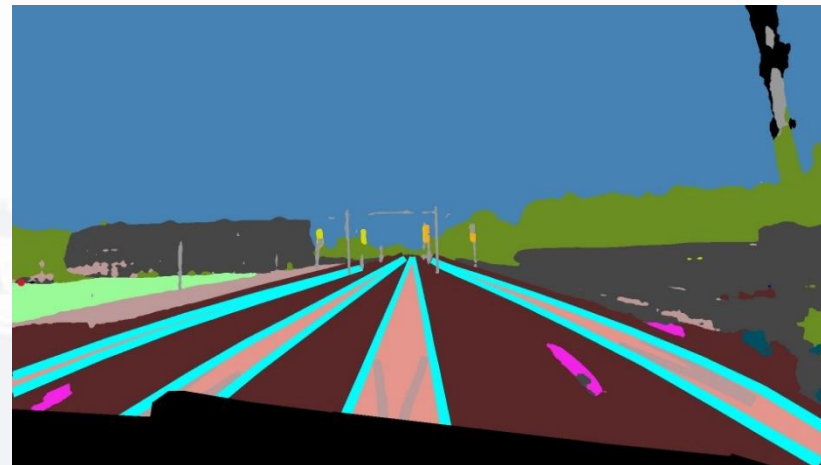


# Pipeline

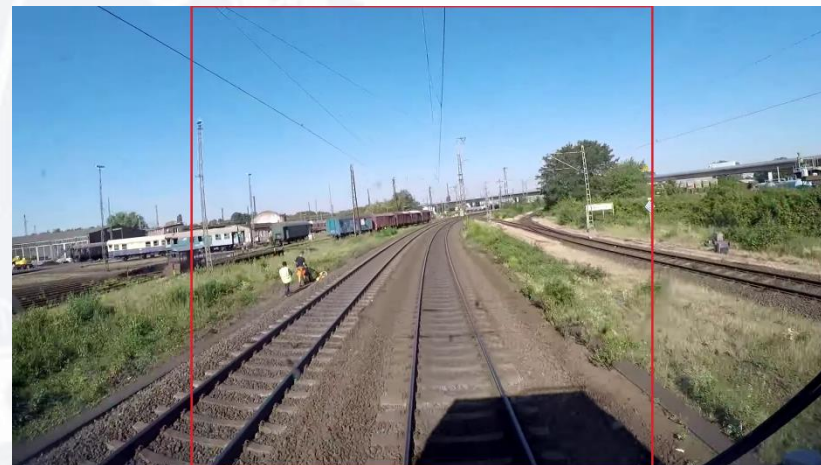


# Experiments

- Dataset: RailSem19
  - 8500 high-resolution images
  - Pre-processing:
    - Middle crop 1080x1080
    - Trained on 512x512
    - Different color schemes for the segmentation masks
    - Additional control (Canny Edge from the original images, etc.)
- BLIP2
  - Network for describing images
  - Improvement over CLIP
  - Usage: prompt extraction



Example semantic segmentation mask



Crop visualisation

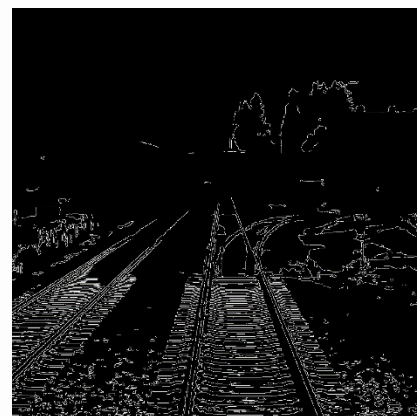
# Experiments

- Control types:
  - Original Masks
  - Canny Edges
  - Combined
- Prompt types:
  - No prompt
  - Fixed prompt  
(*“railway image from the perspective of the train”*)
  - BLIP2 prompt  
(*“a view of a train track with a train on it, high quality, extremely detailed, 4K, HQ”*)
  - Negative prompts  
(*“accidents, derailments, overgrown tracks, missing switches, ...”*)

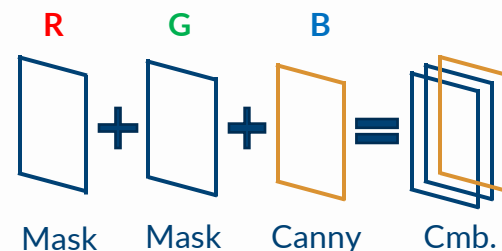
Original  
Mask\*



Canny  
Edges



Combined



# Results

Prediction

Original Image



**Prompt:** “train tracks high-quality image”  
Trained on BLIP2 prompts\*

# Results

Prediction

Original Image



**Prompt:** “train tracks high-quality image”  
Trained on BLIP2 prompts\*

# Results

Prediction

Original Image

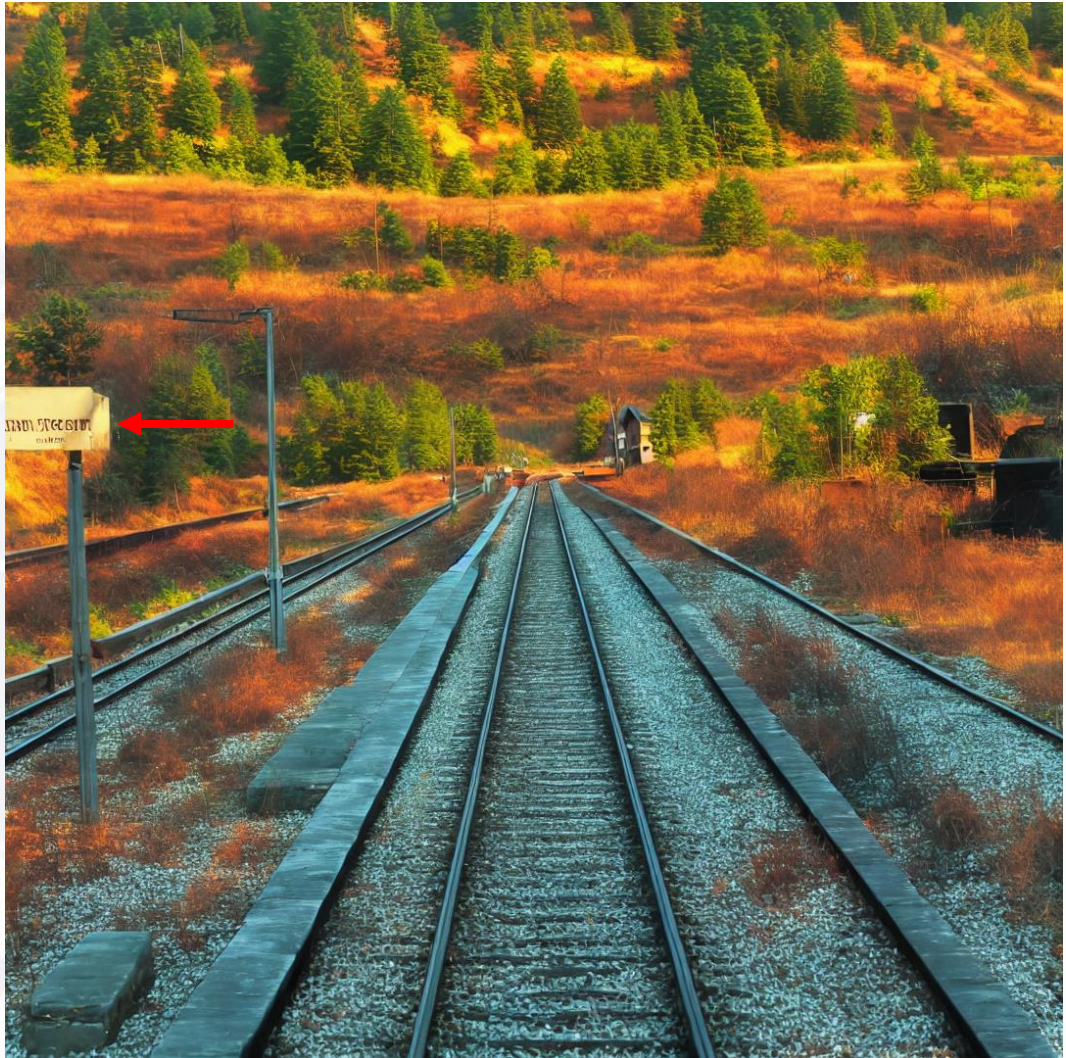
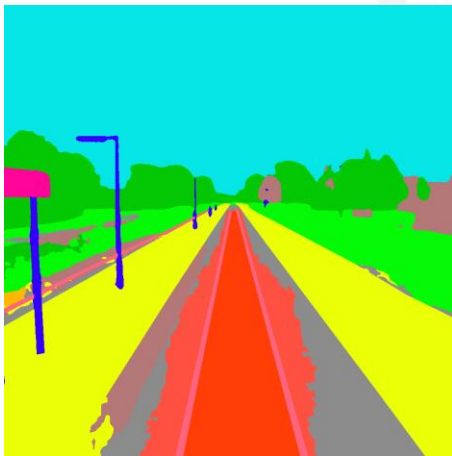


**Prompt:** “a train track with a train on it and a snow covered area, high quality, extremely detailed, 4K, HQ”  
Trained on BLIP2 prompts\*

# Results

Prediction

Original Image

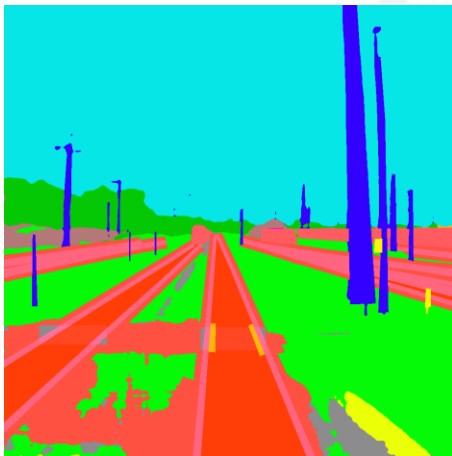


**Prompt:** “train tracks high-quality image”  
Trained on BLIP2 prompts\*

# Results

Prediction

Original Image



Control Image Grayscale\*



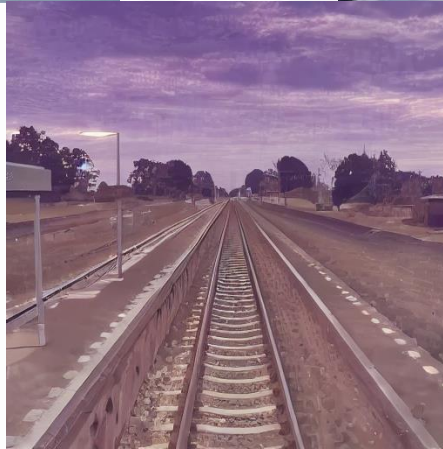
**Prompt:** “a train track with snow on it and wires hanging from the sky,  
high quality, extremely detailed, 4K, HQ”  
Trained on BLIP2 prompts\*

# Results

**Conclusion: Similar performance irrespective of combination (MMC, MCC, MCM)**

- Control type: combined (various)
- Prompt type: various

**Generated Images**



# Results

**Different prompts used with the same model** (trained on BLIP2 prompts)

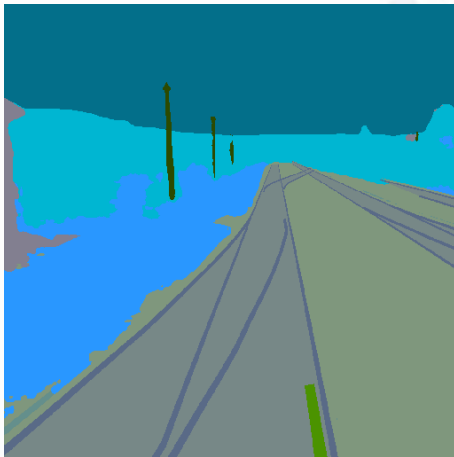
**Original Image**



**No prompt**



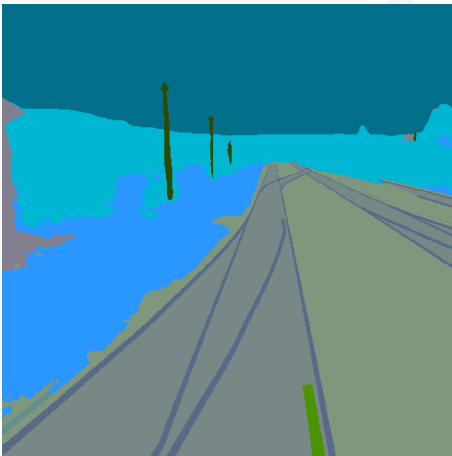
**Prompt:** “a view of a train track with a train on it, high quality, extremely detailed, 4K, HQ”



# Results

How does the latent space affect the generated images?

Original Image



Control Image



Prompt: "train tracks high-quality image"

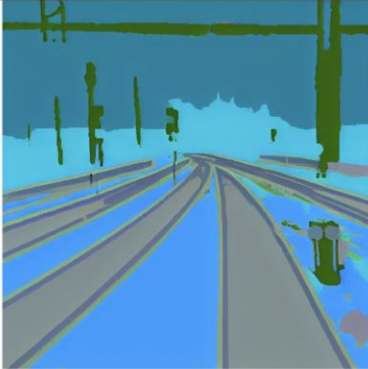
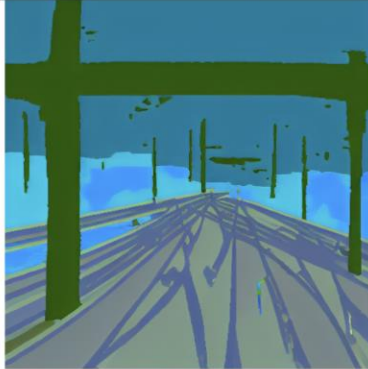
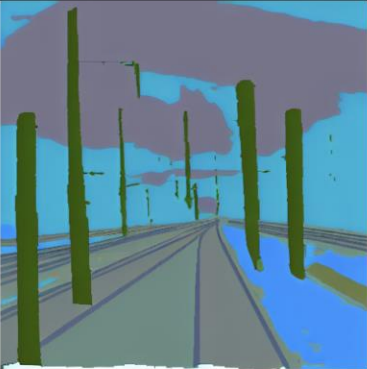
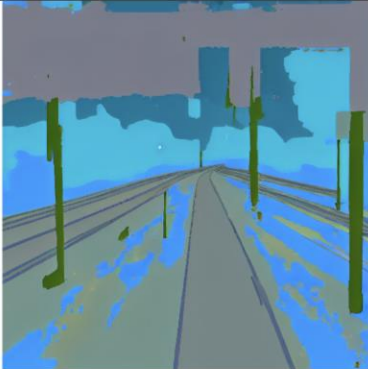
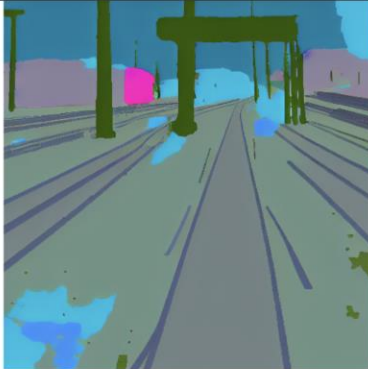
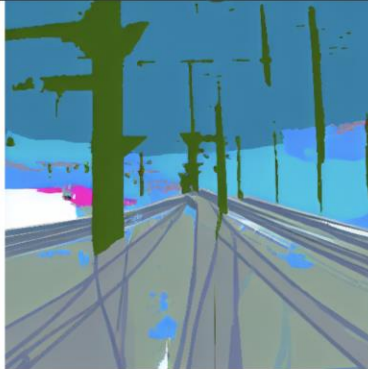
# Results

- The models can accept different resolutions as well (it translates the latent space to images only using Convolutions, not hard constraints there)



# Results

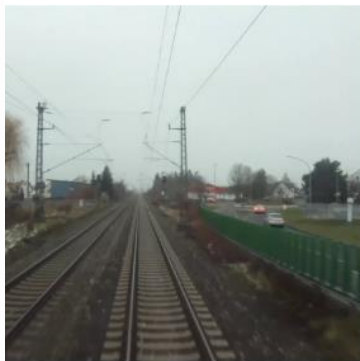
Some segmentation masks generated by InstructorNet.

| Good Results                                                                       |                                                                                    | Average Results                                                                     | Bad Results                                                                          |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <i>"a view of a train track with a few trees in the background"</i>                | <i>"a train track with snow on it"</i>                                             | <i>"a train track with a crossroads on it and a building in the background"</i>     | <i>"a train track with snow on it and wires hanging from the sky"</i>                |
|   |   |   |   |
|  |  |  |  |

# Evaluation – Fréchet Inception Distance

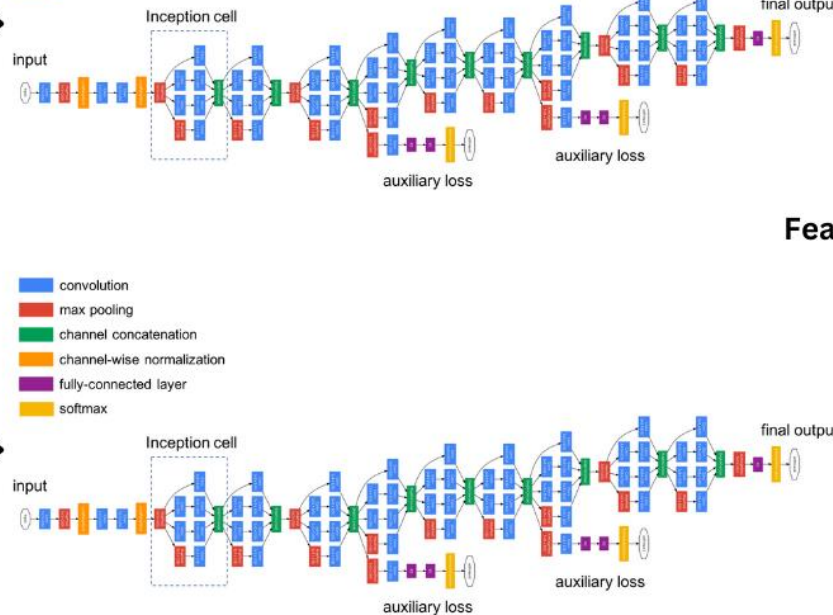


Generated Image

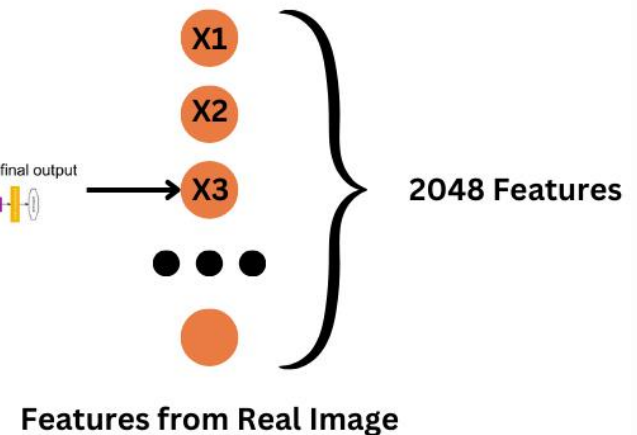
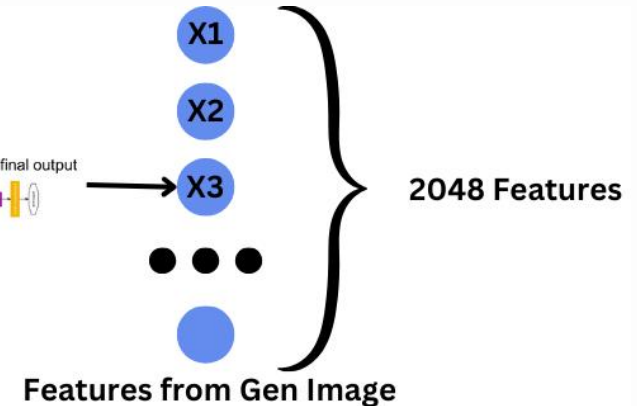


Real Image

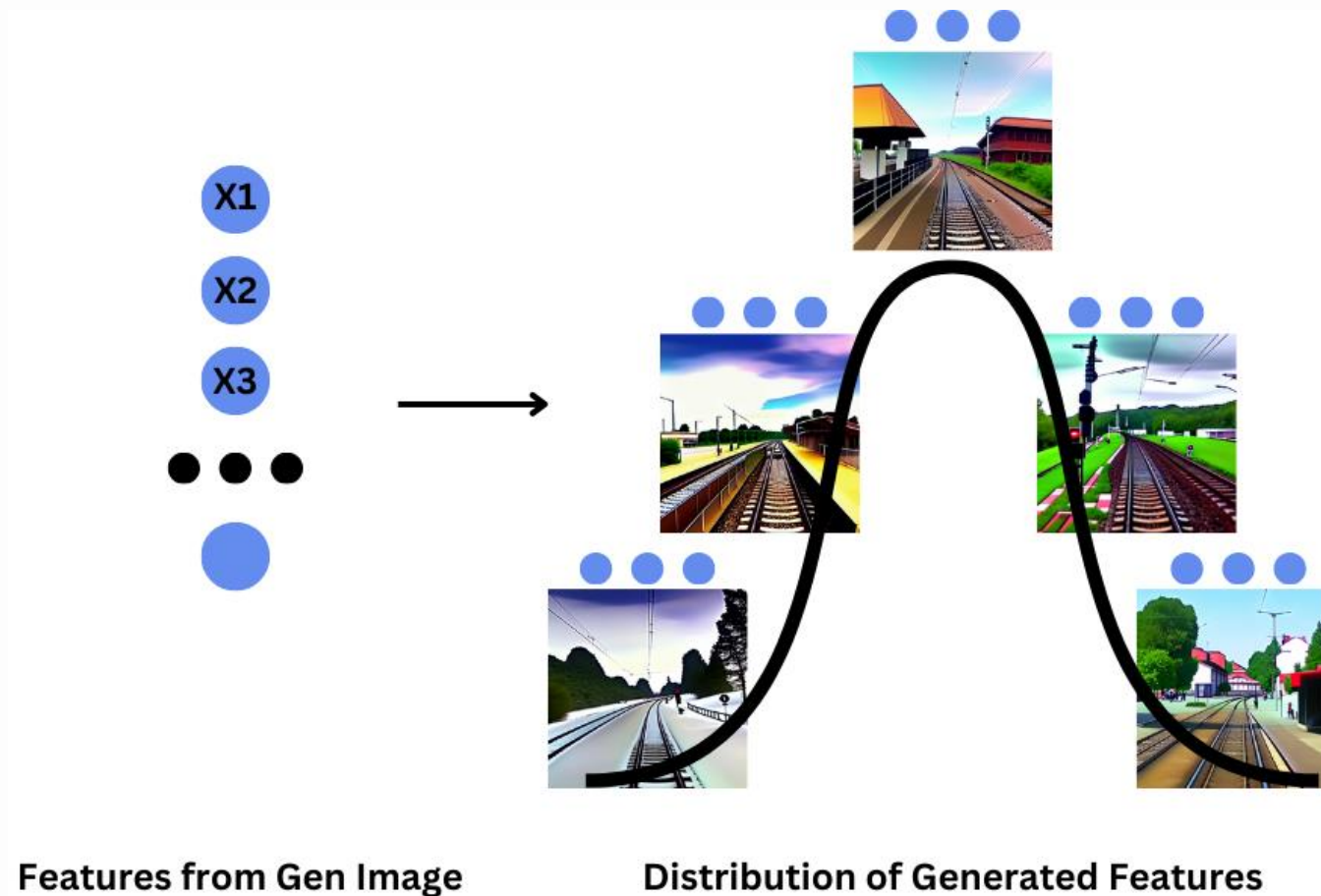
- convolution
- max pooling
- channel concatenation
- channel-wise normalization
- fully-connected layer
- softmax



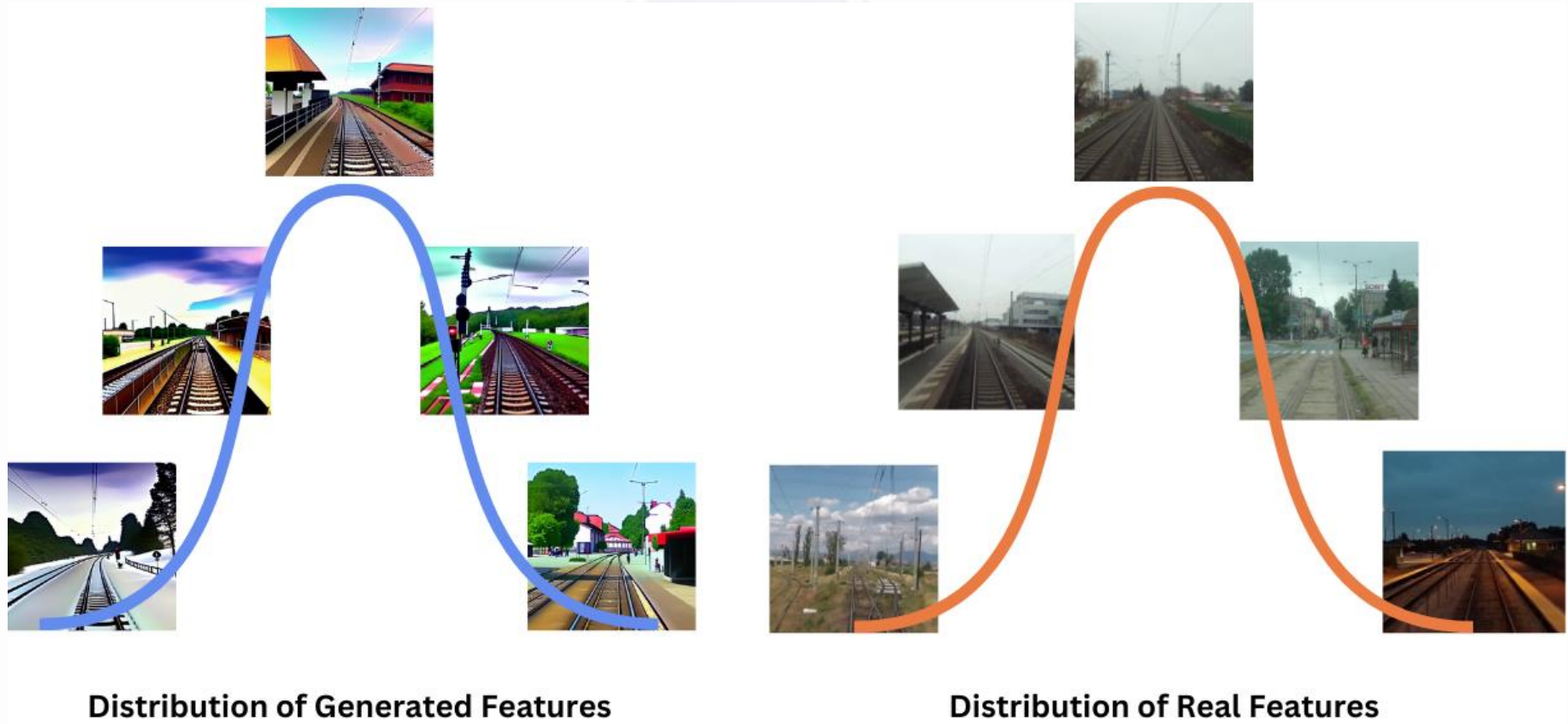
Inception Net V3



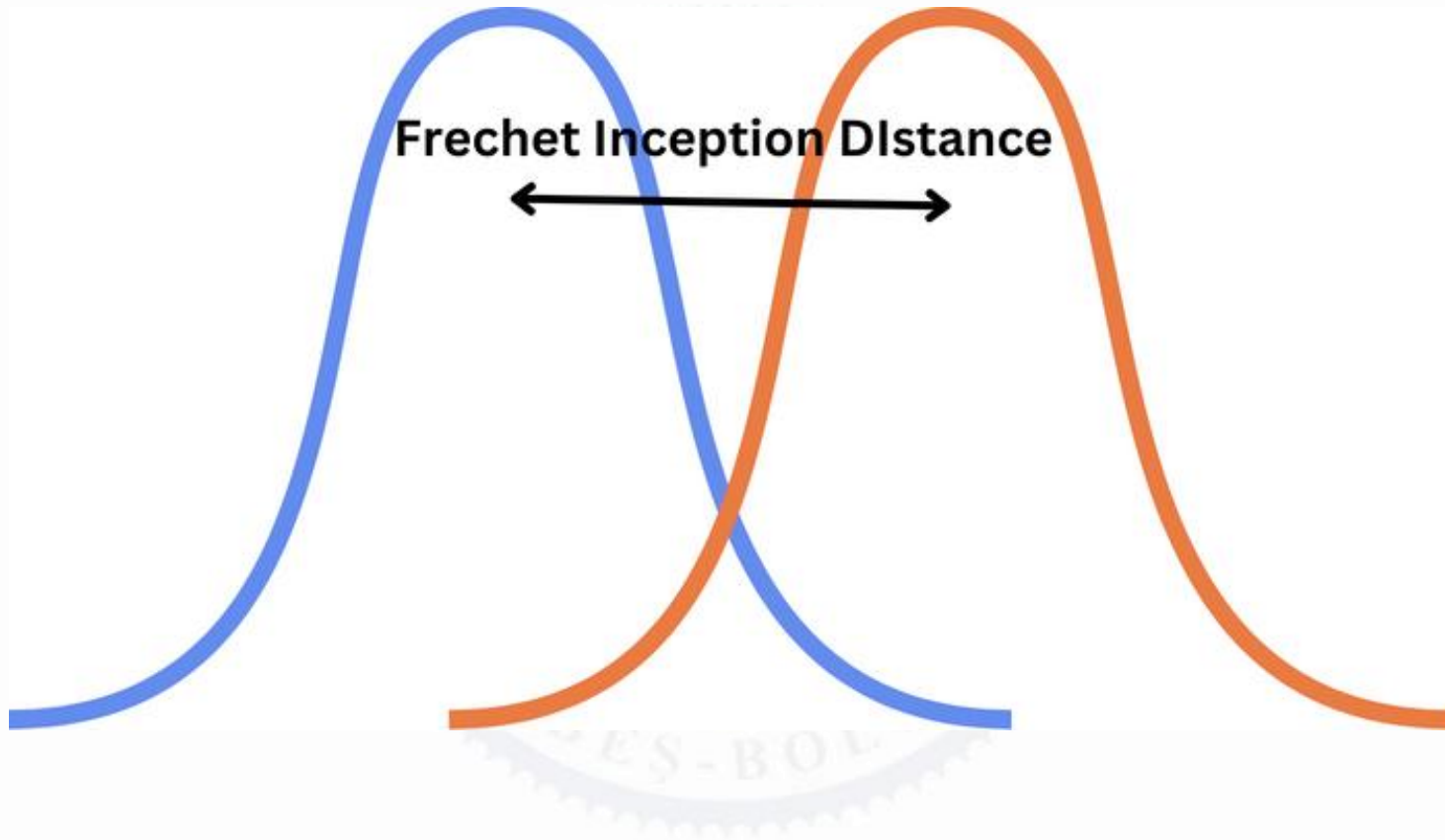
# Evaluation – Fréchet Inception Distance



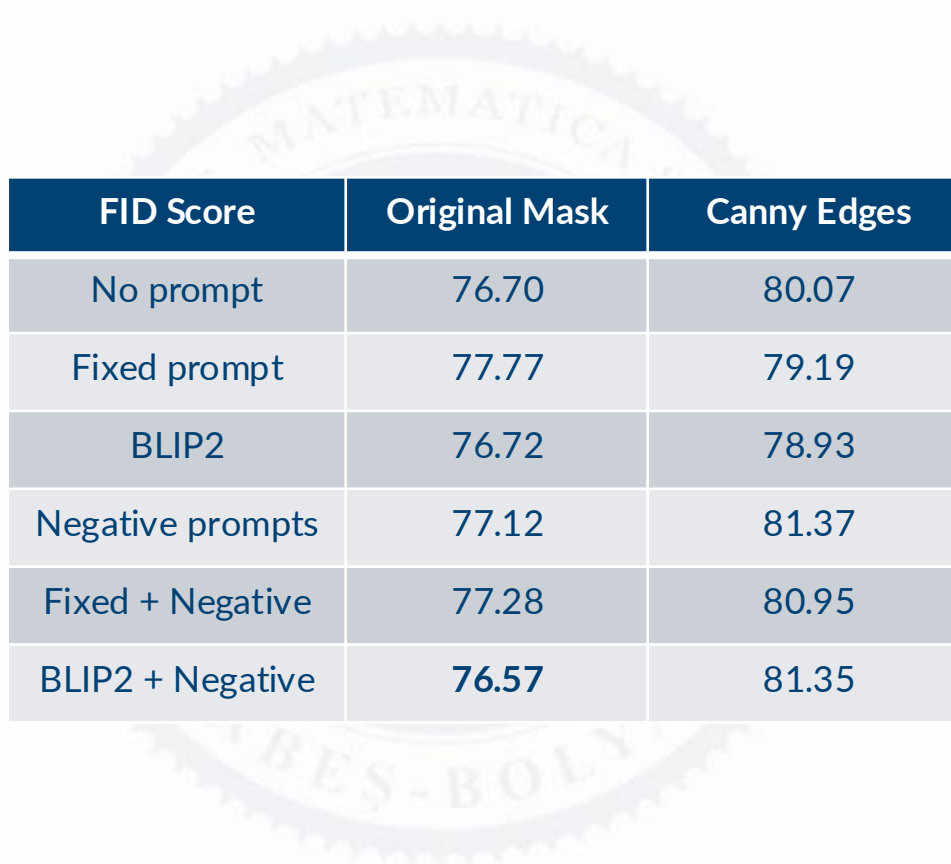
# Evaluation – Fréchet Inception Distance



$$FID = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$



# Evaluation - Fréchet Inception Distance



| FID Score        | Original Mask | Canny Edges |
|------------------|---------------|-------------|
| No prompt        | 76.70         | 80.07       |
| Fixed prompt     | 77.77         | 79.19       |
| BLIP2            | 76.72         | 78.93       |
| Negative prompts | 77.12         | 81.37       |
| Fixed + Negative | 77.28         | 80.95       |
| BLIP2 + Negative | <b>76.57</b>  | 81.35       |

# Evaluation – Training U-NET\* on generated data

Multiple setups were considered:

- Setup A: training set consists of 3000 real images and 500 test images.
- Setup B: 3000 synthetic images were considered and the same 500 test images.
- Setup C: the real and synthetic images were combined, obtaining 6000 train images (3000 real, 3000 synthetic), maintaining the same 500 images for testing.
- Setup D: like setup C, with the mention that only 300 images are selected in total for training, 1500 real and 1500 synthetic.
- Setup E: like setup A, but instead of 3000 training images there are 6000.

| Setup | mIoU    | Std. dev |
|-------|---------|----------|
| A     | 78.493% | 0.293    |
| B     | 78.295% | 0.546    |
| C     | 80.208% | 0.542    |
| D     | 78.369% | 0.253    |
| E     | 80.268% | 0.686    |

\* The model was trained on binary segmentation, retaining only the rails (so a class for rails and another for non-rails).

# Applications\*

Some applications of the proposed pipeline would be:

- **Autonomous Trains Development:** Those synthetic pairs of image-mask generated from text descriptions can be used to train and test semantic segmentation models for autonomous vehicles. This enables the development and validation of perception systems in a controlled environment before real-world deployment, enhancing safety and reliability.
- **Railway Maintenance and Inspection:** Semantic segmentation models trained on synthetic images can assist in the automatic detection and classification of defects or anomalies in railway infrastructure, such as cracks in tracks, damaged signals, or vegetation encroachment. This can streamline maintenance operations and improve overall safety.
- **Training Simulators for Railway Operators:** Synthetic rail scene images can be utilized to develop training simulators for railway operators and personnel. By providing realistic scenarios, operators can practice various tasks such as signal recognition, obstacle detection, and emergency response in a safe and controlled virtual environment.

\* This section was made using Gemini (from Google).

# SWOT Analysis\*

## Strengths:

- **High-Quality Image Generation:** Text-to-image diffusion models are known for their ability to create realistic and diverse images. This can be beneficial for generating a large and varied dataset of rail scene images for semantic segmentation tasks.
- **Open-Ended Scene Description:** Text prompts allow for detailed descriptions of the scene, including specific objects, weather conditions, and lighting. This flexibility can lead to a more comprehensive dataset compared to traditional methods.
- **Potential for Unseen Categories:** The model might be able to handle object categories not present in the training data, due to the reliance on textual descriptions.

## Weaknesses:

- **Accuracy and Control:** Diffusion models might struggle with precise segmentation, especially for small objects or complex scenes. Text prompts may not fully capture the intricacies needed for accurate segmentation.
- **Limited Training Data Availability:** Training text-to-image diffusion models requires a vast amount of data. Specialized datasets for rail scenes with corresponding text descriptions may be scarce.
- **Computational Cost:** Training and running diffusion models can be computationally expensive, requiring powerful GPUs.

\* This section was made using Gemini (from Google).

# SWOT Analysis\*

## Opportunities:

- **Focus on Safety-Critical Objects:** Semantic segmentation of rail scenes can be crucial for applications like automated train inspection, focusing on accurate segmentation of tracks, signals, and other safety-critical objects.
- **Domain-Specific Training:** Fine-tuning the diffusion model on existing, labeled rail scene datasets can improve its performance in this specific domain.
- **Exploration of Alternative Techniques:** Techniques like combining diffusion models with traditional segmentation methods might lead to better segmentation accuracy.

## Threats:

- **Evolving Regulations:** Regulations around data privacy and security in the rail industry might limit data collection for training the model.
- **Alternative Approaches:** Other techniques for semantic segmentation, such as encoder-decoder architectures, might prove more efficient or accurate for this specific task.

\* This section was made using Gemini (from Google).

# Conclusion

- Generated realistic railway images
- Best combination: original masks + BLIP2 prompts
- Negative prompts lead to small improvements

# Future Work

- Better mappings for some classes (sky, signs)
- More research for InstructorNet, to explore its limits



**Thank you for your attention!**

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